

A Counterexample to DeVos’s Average Reachability Conjecture

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Abstract

We give two explicit counterexamples to a natural average reachability formulation of DeVos Conjecture 6.10. The underlying objects are directed graphs whose arcs carry labels, and a walk may use those labels only in increasing order. For a uniformly chosen starting vertex, a five-vertex graph with two labels has average reachability $19/5$, below the proposed lower bound 4. We then treat a more restrictive variant: it forbids a rainbow directed cycle in every label order and averages over both starting vertices and label orders. A nine-vertex example still has average $62/9 < 7$. Both finite arguments are verified in Lean 4.2.4 using exact rational arithmetic.

1 Introduction

DeVos Conjecture 6.10, as recorded by Sullivan [1], concerns reachability in directed graphs with labelled arcs. The formulation studied here asserts that

$$\frac{1}{n} \sum_{v=0}^{n-1} |\text{Reach}_{(0,1,\dots,k-1)}(v)| \geq 1 + \sum_{i=0}^{k-1} \min_v d_i(v).$$

The starting vertex counts as reachable, which accounts for the initial 1 in the proposed bound.

Under this interpretation, the conjecture fails. The first example is small enough to inspect directly and already disproves the uniform-start statement. One might hope that the issue is merely the chosen order of the labels. The second example rules out that repair as well: it satisfies the relevant no-rainbow-cycle condition in every label order, yet violates the corresponding order-averaged inequality.

There is no large search hidden in either construction. The graphs are explicit, and the accompanying Lean development checks every finite claim. We include the definitions and exact certificates so that the counterexamples can be verified without relying on numerical experimentation or informal case checking.

2 Ordered labelled digraphs

Fix positive integers n and k . An n -vertex, k -labelled digraph is a function

$$E : \{0, \dots, k-1\} \times \{0, \dots, n-1\}^2 \longrightarrow \{0, 1\}.$$

We write $E_i(v, w) = 1$ when the directed arc $v \rightarrow w$ carries label i . Different labels may occur on the same ordered pair, and opposite arcs are allowed.

Given an ordered list $I = (i_1, \dots, i_r)$ of distinct labels, define $\text{Reach}_I(v)$ recursively. The walk may skip a label or use one outgoing arc of that label and then continue with the remaining labels:

$$\begin{aligned} \text{Reach}_\emptyset(v) &= \{v\}, \\ \text{Reach}_{(i)::I}(v) &= \text{Reach}_I(v) \cup \bigcup_{E_i(v,u)=1} \text{Reach}_I(u). \end{aligned}$$

In other words, the labels used by a walk are strictly increasing in the selected order. For the canonical order $I_k = (0, 1, \dots, k-1)$, put

$$r_E(v) = |\text{Reach}_{I_k}(v)|, \quad \bar{r}(E) = \frac{1}{n} \sum_{v=0}^{n-1} r_E(v).$$

Definition 2.1 (Increasing-cycle condition). The labelled digraph E is *increasing-cycle-free* if there is no nonempty closed walk whose labels are strictly increasing in the canonical order. Equivalently, the recursive reachability computation has no nontrivial closed reach from any vertex.

Let $d_i(v)$ be the outdegree of v in layer i , and let

$$m_i(E) = \min_v d_i(v), \quad B(E) = 1 + \sum_{i=0}^{k-1} m_i(E).$$

The uniform-start conjecture is the assertion

$$\bar{r}(E) \geq B(E) \tag{C}$$

for every n, k with $n > 0$ and every increasing-cycle-free E .

3 A five-vertex counterexample

Take $n = 5$, $k = 2$, and vertices $0, 1, 2, 3, 4$. The outgoing arcs in the two layers are listed in Table 1. An omitted arc is absent.

Table 1: The five-vertex counterexample.

vertex v	E_0 -successors	E_1 -successors
0	1,4	4
1	0,3	4
2	1,4	1
3	0,4	0
4	2,3	1

Every vertex has layer-0 outdegree 2 and layer-1 outdegree 1; hence $B(E) = 1 + 2 + 1 = 4$. The reachable sets in canonical order are

v	0	1	2	3	4
$\text{Reach}_{(0,1)}(v)$	$\{0, 1, 4\}$	$\{0, 1, 3, 4\}$	$\{1, 2, 4\}$	$\{0, 1, 3, 4\}$	$\{0, 1, 2, 3, 4\}$.

Their sizes sum to 19, so $\bar{r}(E) = 19/5 < 4$.

Theorem 3.1. *The uniform-start conjecture (C) is false.*

Proof. The graph in Table 1 is increasing-cycle-free. Because there are only two labels, a nonempty increasing closed walk would have to contain either a loop or a two-edge rainbow directed cycle. The finite verification rules out both possibilities. The minimum outdegrees and reachable sets computed above give

$$\bar{r}(E) = \frac{19}{5} < 4 = B(E),$$

The exact finite verification is the Lean theorem `VertexAverageConjecture_false`.

□

4 A label-order-averaged variant

For an arbitrary permutation π of the k labels, define $\text{Reach}_\pi(v)$ by the same recursion. Consider the condition that no nonempty increasing closed walk exists for *any* permutation π . Define the order-averaged reachability by

$$\bar{r}_{\text{ord}}(E) = \frac{1}{n k!} \sum_{\pi \in S_k} \sum_{v=0}^{n-1} |\text{Reach}_\pi(v)|.$$

The associated proposed inequality is

$$\bar{r}_{\text{ord}}(E) \geq B(E). \quad (\text{C}')$$

The next construction refutes (C') even with the stronger all-orders cycle condition. Take $n = 9$, $k = 2$, and use identical layers. The common layer has successor sets

v	successors in each layer
0	1, 3, 4
1	2, 3, 4
2	0, 3, 4
3, 4	5, 6, 7
5, 6, 7, 8	0, 1, 2.

The common layer is loopless and oriented, so it has no directed cycles of length one or two. With only two labels, that is exactly what is needed to exclude a rainbow directed cycle in either order.

In both orders $(0, 1)$ and $(1, 0)$, the reachable-set sizes, listed by starting vertex, are

$$(8, 8, 8, 7, 7, 6, 6, 6, 6).$$

Thus the total over the two orders is 124, and

$$\bar{r}_{\text{ord}}(E) = \frac{124}{9 \cdot 2} = \frac{62}{9} < 7.$$

Each vertex has outdegree 3 in each layer, so $B(E) = 1 + 3 + 3 = 7$.

Theorem 4.1. *The all-orders averaged conjecture (C') is false.*

Proof. The construction satisfies the no-rainbow-cycle condition in both orders of the two labels. Its minimum outdegrees are $(3, 3)$, while its exact order-averaged reachability is $62/9 < 7$. The complete finite argument is checked by the Lean theorem

`VertexAverageConjectureWeak_false.`

□

5 Formal verification

The Lean development uses no floating-point computation. The graph model, recursive reachability functions, exact rational averages, and both conjecture statements are defined in `Definitions.lean`. The first construction and its certificates are in `devosProofStrong.lean`; the second is in `devosProofWeak.lean`; and the two global negation theorems are in `devosNegation.lean`.

All displayed finite equalities are established by Lean’s kernel-checked decision procedure (the proofs use `by decide`). In particular, the formal development verifies the relevant cycle exclusions, every outdegree and minimum outdegree, every reachable-set cardinality, both exact averages, and the strict inequalities that yield the contradictions. The project builds as the Lean package `devosOrderedLabel` with Lean 4.2.4.

6 Conclusion

The proposed lower bound fails already with two labels. The five-vertex graph disproves the uniform-start formulation. The nine-vertex graph shows that averaging over label orders does not restore the inequality, even if one requires the absence of a rainbow directed cycle in every order. Since both arguments are checked with exact rational arithmetic, the comparisons contain no rounding issue.

References

- [1] J. Sullivan, *A Summary of Results and Problems Related to the Caccetta–Haggkvist Conjecture*, 2006, Conjecture 6.10, p. 7.